

A QGIS Analysis of Chinese Belt and Road Initiative Maritime Structure Investment in Eastern Africa

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Introduction:

Eastern Africa is the most populous region on the continent, hosting countries such as Ethiopia, Kenya, Tanzania, and more. Not only is it incredibly populated, but the eastern shore of Africa hosts some of the most geostrategically important points in the world, such as the Bab al-Mandab Strait. Also, the eastern shore has rivers that bring critical resources from Asia to landlocked countries like Uganda, Rwanda, and Burundi. Nations from all around the world understand the significance of economic and humanitarian development of the continent which is predicted to become the world's most populated by the end of the century. China has swooped in to be one of the first major investors in Eastern African infrastructural development, but their motives are unclear. Different schools of academic thought have differing opinions on the matter, but an undeniable fact is that the Belt and Road Initiative has been a driving force in why China has focused so heavily on this region.

There is plenty of debate on whether China is an imperialist country. Since the end of the First Industrial Revolution around the mid 19th century, China has not made any major land grabs, so it does not fit under the umbrella of European-style imperialism. However, there is obvious tension between neorealist thinkers and more liberal viewpoints. To offensive neorealists, China is nearly a token example of power projection from a hegemonic nation that continues to rise in power. The argument is simple: since the international system is anarchic, China naturally expands its sphere of influence to ensure its own security, only in this example, China builds its empire through close economic ties with nations who seek to develop quickly and cheaply in the short term. Liberal thinkers see it differently. They believe that China had to create BRI out of necessity due to domestic issues stemming from overproduction of steel and other manufacturing resources. To keep their economy from crashing due to the extreme surplus of goods, China had

no other option but to turn towards exporting those goods rather than shut down enough factories to cause a serious unemployment crisis, and thus BRI became Xi Jinping's signature policy.

Regardless of perspective, GIS is essential to visualize the current situation of BRI in Eastern Africa. I have chosen three of what I believe to be the most impactful ports in the region that BRI funded or built, all within the last ten years. First, completed in 2017 by the Chinese Harbor Engineering Company, was the Doraleh Multipurpose Port in Djibouti (Djibouti Ports & Free Zones Authority, 2019). Next, in 2021, was the modernization Lamu Port in Kenya, completed by China Merchants Port Holdings Company Limited (Kenya Ports Authority, 2026). Finally, the China Communications Construction Community was responsible for the massive modernization and expansion of the Port of Dar es Salaam in Tanzania in 2023 (Tanzania Ports Authority, 2026). There are other Chinese-owned companies that have already won contracts for new ports further expansions to existing ports within the region, but I settled on three completed projects. This project argues that QGIS analysis reveals a pattern of Chinese investment in strategically significant maritime infrastructure. While geospatial analysis cannot determine political intentions behind these investments, it provides a framework for measuring scale and strategic positioning of infrastructure that is central to debates surrounding BRI.

Objective:

The objective of this project was to create a geospatial dataset of three strategically significant East African ports associated with BRI through manual digitization of satellite imagery. The project also compares certain physical characteristics of each port to demonstrate the application of GIS techniques in geopolitical infrastructure analysis.

Data Collection:

I used Google Satellite imagery in QGIS as a base layer for manual digitization. I also sourced publicly available information to determine port surface area (square meters), perimeter (meters), number of births, build cost (USD), depth (meters), finish year, and port freight traffic (tonnes). All the details for these attributes come from official port reports and websites. Together, those details help paint a picture on how a port is equipped to handle logistical details such as container size, type of freight, loading/unloading goods, and more.

Methodology:

I created multiple shapefile layers for project data and manually digitized each layer using QGIS polygons based on Google Satellite imagery, simply digitizing with straight line segments. Then, using derivations from within QGIS, I calculated the area and perimeter of each port to determine operational footprint and assess land-use efficiency, given that the Port of Dar es Salaam is directly within the city while the other two lie in less densely built areas. Finally, I used Python to clean, organize, and categorize attribute data and then I produced the final cartographic layouts seen below.

Results:

Figure 1:
Overview

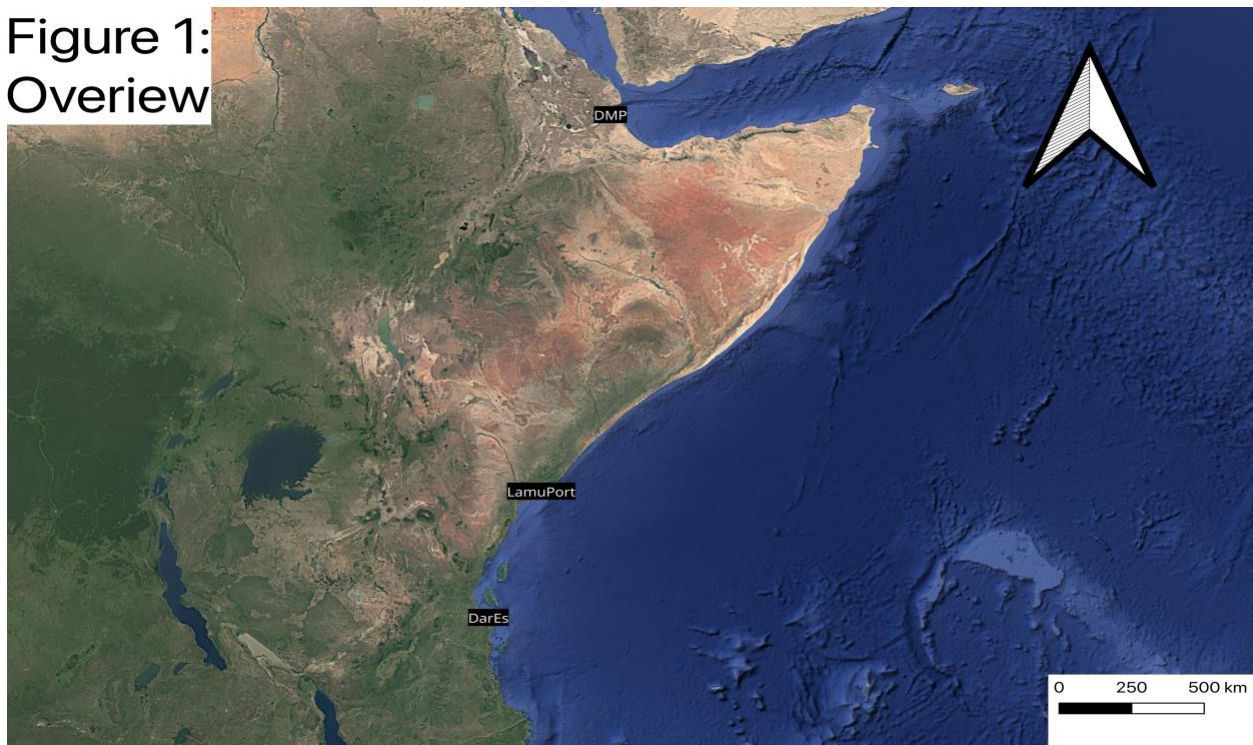


Figure 1 shows the greater Eastern Africa without borders to help readers understand the scale of Chinese involvement.

Figure 2: Port of
Dar es Salaam

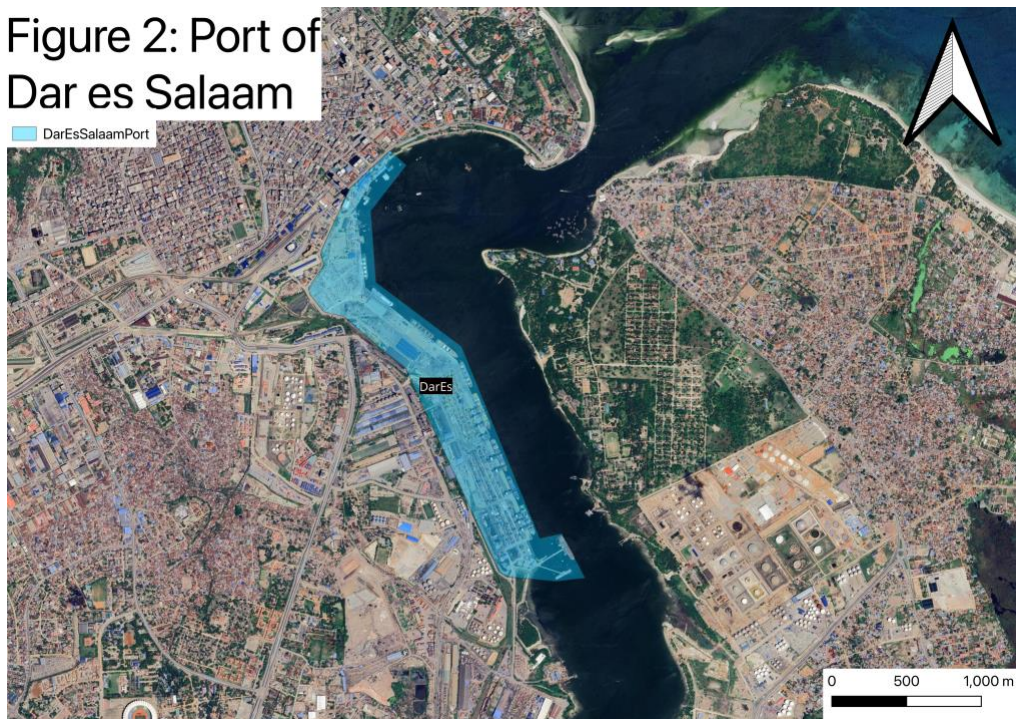


Figure 2 shows the more elongated Port of Dar es Salaam. Its unique shape second in terms of perimeter and last in area, its 12 berths give it an overwhelming advantage in total freight capacity and make it the most efficient in terms of traffic per meter. Its biggest advantage is arguably simply being in the heart of Dar es Salaam, Tanzania's largest city and one of the fastest-growing cities in the world. The already established economy gives it a strong base for skilled labor and logisticians.

Figure 3: Doraleh Multi-purpose Port



Figure 3 shows the more uniform DMP. The leftmost jetty and the compound that it leads to is the only permanent Chinese military base in the region. There are plans to expand to 16-17 berths total.

Figure 4: Lamu Port

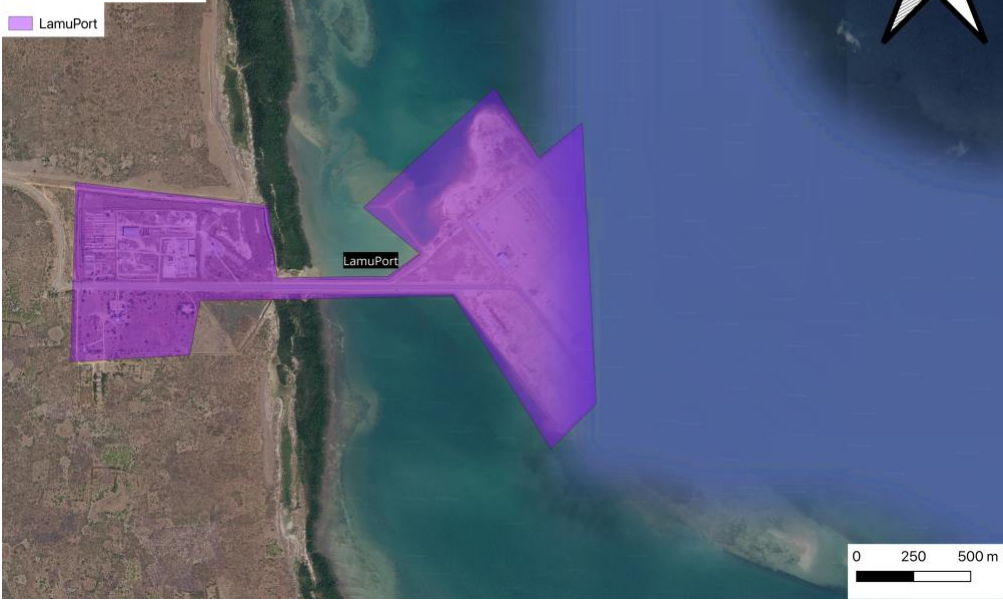


Figure 4 unfortunately evidences limitations of satellite imagery. Official open-source data suggests that the offshore terminal is rectangular in shape, so while one can estimate the dimensions, the satellite imagery shows no proof. Notwithstanding the lack of imagery, Lamu Port is largest in both area and perimeter despite only having 3 berths. There future are plans for massive expansions of up to 8 kilometers in area and 23 to 32 berths in total.

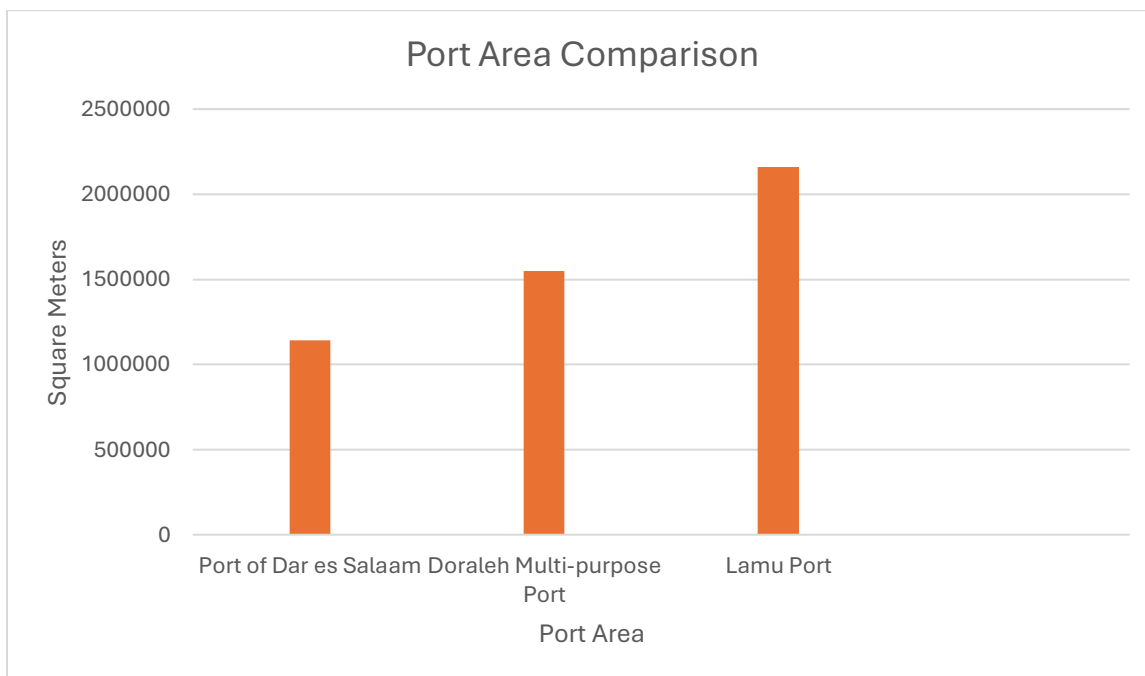
Table 1: Geospatial Characteristics

	Country	Area (sqm)	Perimeter (m)
Dar es Salaam Port	Tanzania	1,143,486	7,632
Doraleh Multi-purpose Port	Djibouti	1,548,272	5,122
Lamu Port	Kenya	2,161,245	9,602

Table 2: Infrastructural Characteristics

	Berths	Depth (m)	Build Cost (\$)	Finish Year	2025 Tonnes	BRI Partner(s)
Dar es Salaam	12	14.5	421,000,000	2023	27,700,000	CHEC
Doraleh	6	15.3	590,000,000	2017	3,400,000	CMPort
Lamu	3	18	470,000,000	2021	800,000	CCCC

Graph 1: Geospatial Characteristics



Discussion:

The primary objective of this study was to use satellite imagery via QGIS to determine if Chinese BRI-related port investments in Eastern Africa were solely economic in nature or if there were geostrategic intentions behind the investments. First, the Port of Dar es Salaam is in the heart of the most populous city and second largest economy (Uchumi, 2026) in the region. Already a major transportation hub, the massive modernization of the port gave China greater

control over transportation within not only Tanzania, but also inland countries that rely on rivers that run to the Indian Ocean via Tanzania. Next, the Doraleh Multi-purpose Port is directly adjacent to the only permanent Chinese military base in the greater Middle East. Built a short distance away from Djibouti City, the port sits to the south of one of the narrowest points of the Bab al-Mandab Strait, a major oil transportation point. Finally, Lamu Port is majorly important within China's greater web of port investments throughout the Indian Ocean. Since it is in an early phase of greater planned construction, the current 3-berth port serves as a first step through the door in Kenya and it is a reminder that China has farsighted geostrategic ambitions.

Altogether, China financed over \$1 billion which created strategic dependence through interest and debt payments and developed a wider sphere of influence throughout Eastern Africa and the Indian Ocean.

Limitations:

The Google Satellite imagery could not capture the operational area of Lamu Port. Therefore, neither the resulting polygon nor the measured attributes fully represent the entirety of the port. Importantly, because measurements were taken manually, the aim of this study is not to determine polygon borders with 100% accuracy, but to use satellite imagery to achieve 85% accuracy. Also, this case is limited to only the three ports studied and is not meant to represent all BRI-related ports in Africa.

Conclusion:

In conclusion, this study used geospatial analysis to evaluate geospatial and infrastructural characteristics of Eastern African ports related to BRI with the goal of recognizing patterns in Chinese investments that exhibit geostrategic motives more than economic. Ultimately, the data

suggests that BRI-related ports were built to suit China's geostrategic interests in the region. As the listed projects continue into new phases and new projects start, China will continue to make moves that enhance their position in the region disguised as economic interests. The infrastructural aspect is only one part of the greater Belt and Road Initiative and the future will bring China continuing to tie together infrastructure with mining, technology, and more.

Future Work: This project could be expanded to incorporate shipping lanes or further BRI projects within the region, specifically railroads. Further analysis of the greater infrastructure system within Eastern Africa would be incredibly valuable for greater growth within the region. The region is the most populous on the continent and understanding the logistics of resource distribution is key to economic and humanitarian development in the region and Africa as a whole. Not only that but building upon the ideas developed in this project would lead to more accurate analysis of China's involvement within the region and would shed more light within the scope of the project on whether China is looking to expand its empire through debt trapping the region or if it simply is building out of economic necessity. For example, a further project would seek to interpret China's end goal in Djibouti and the Bab al-Mandab Strait, given that the Doraleh Multi-purpose Port is directly adjacent to China's only permanent base within the greater East Africa/Middle East region.

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